

Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: nabil.ahmed@bracu.ac.bd

Table of Contents

1 Some Special Continuous Probability Distributions

- Exponential Distribution
- Normal Distribution

CDF

- The CDF of a continuous r.v. X can be written as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$$

- For example, if $x = a$:

$$F(a) = P(X \leq a) = P(X < a) = \int_{-\infty}^a f(x)dx$$

- **Note:** For a continuous r.v. X , $P(X = a) = 0$

Probability from CDF

$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a)$$

- $P(X \leq a) = F(a)$
- $P(X > a) = 1 - P(X \leq a) = 1 - F(a)$

Exponential Distribution

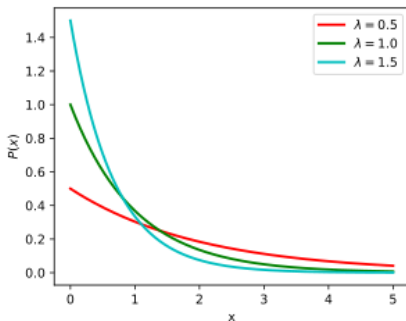
- If the pdf of a continuous r.v. X is given by

$$f(x) = \lambda e^{-\lambda x}; x > 0, \lambda > 0$$

- then X is said to have an exponential distribution where $\lambda > 0$ is the rate parameter
- It is a valid pdf:

$$\begin{aligned}\int_0^{\infty} f(x) dx &= \int_0^{\infty} \lambda e^{-\lambda x} dx \\&= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{\infty} \\&= -1(e^{-\infty} - e^0) \\&= -1(0 - 1) \\&= 1\end{aligned}$$

Shape of the Exponential Distribution



CDF of an Exponential R.V.

- CDF of an exponential r.v. X :

$$\begin{aligned}F(x) &= P(X \leq x) \\&= \int_0^x f(x) dx \\&= \int_0^x \lambda e^{-\lambda x} dx \\&= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^x \\&= -1(e^{-\lambda x} - 1) \\&= 1 - e^{-\lambda x}\end{aligned}$$

- Therefore, for an exponential r.v. X , $F(x) = 1 - e^{-\lambda x}$

Example

- Suppose that the length of a phone call in minutes is an exponential r.v. with parameter $\lambda = 0.1$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you have to wait
 - (a) more than 10 minutes.
 - (b) between 10 and 20 minutes.

Example (Solution)

- Let X denote the length of the call made by the person in the booth.

(a)

$$\begin{aligned}P(X > 10) &= 1 - F(10) \\&= 1 - [1 - e^{-0.1 \times 10}] \\&= e^{-1} \\&= 0.368\end{aligned}$$

(b)

$$\begin{aligned}P(10 \leq X \leq 20) &= F(20) - F(10) \\&= [1 - e^{-0.1 \times 20}] - [1 - e^{-0.1 \times 10}] \\&= e^{-1} - e^{-2} \\&= 0.233\end{aligned}$$

Mean and Variance of an Exponential R.V.

- Show that the mean and variance of an exponential r.v. X are $E(X) = \frac{1}{\lambda}$ and $V(X) = \frac{1}{\lambda^2}$, respectively.
- Prove that the mean and standard deviation of an exponential r.v. are the same.

Exercise

- The time (in hours) required to repair a machine is an exponentially distributed r.v. with parameter $\lambda = 0.5$. Find
 - (a) the probability that a repair time exceeds 2 hours.
 - (b) the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours.
 - (c) the expected repair time.

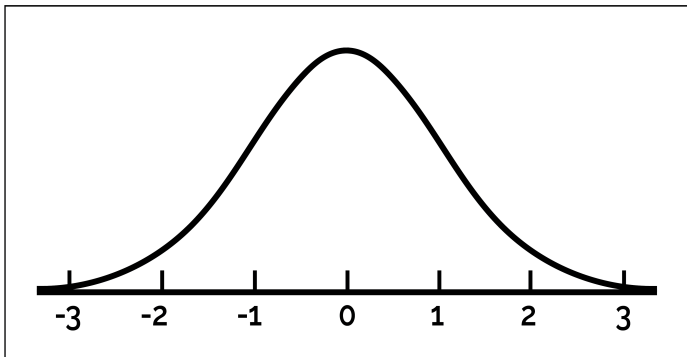
Normal Distribution

- For a continuous r.v. X , if the pdf is given by

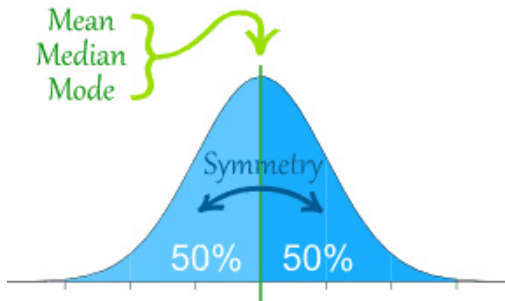
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma^2 > 0$$

- then X is said to have a normal distribution with μ and σ parameters
- Properties:
 - the distribution is bell-shaped
 - it has two parameters (μ is the mean and σ^2 is the variance)
 - It is symmetric about mean μ

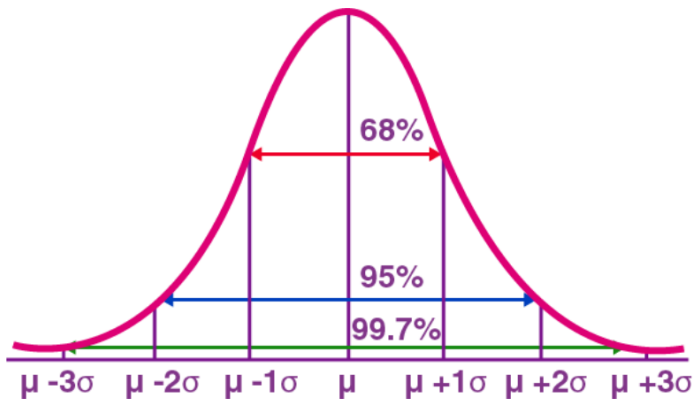
Shape of a Normal Distribution



Shape of a Normal Distribution



Shape of a Normal Distribution



Mean and Variance of a Normal R.V.

- The mean and variance of a normal r.v. X are $E(X) = \mu$ and $V(X) = \sigma^2$, respectively.

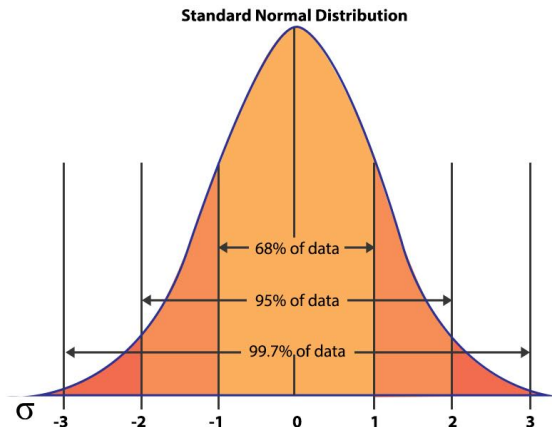
Standard Normal Distribution

- If a normal distribution has a mean $\mu = 1$ and variance $\sigma^2 = 1$, then we call this a standard normal distribution.
- That is, if $Z \sim N(0, 1)$, then the pdf of Z can be written as

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; -\infty < z < \infty$$

- **Exercise:** Show that the pdf of this standard normal distribution is a valid pdf.

Shape of a Standard Normal Distribution



CDF of a Standard Normal R.V.

- The CDF of a standard normal r.v. Z can be written as

$$F(z) = P(Z \leq z) = \int_{-\infty}^z f(z) dz = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz = \Phi(z)$$

- Here, $\Phi(\cdot)$ is called the CDF of the standard normal distribution. Its value can be directly obtained from the [standard normal table](#) (also called as z table).

Standard Normal Table

T-2 Tables

Table entry for z is the area under the standard normal curve to the left of z .

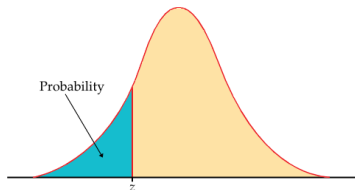


TABLE A

Standard normal probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048

Standard Normal Table

Tables | T-3

Table entry for z is the area under the standard normal curve to the left of z .

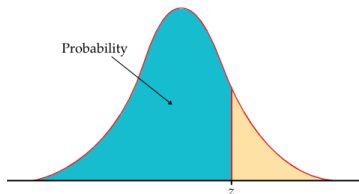


TABLE A

Standard normal probabilities (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224

Exercise

- If X is a normal r.v. with mean μ and variance σ^2 , then show that $Z = \frac{X - \mu}{\sigma}$ follows a standard normal distribution.

Example

- Suppose that the heights of 300 students are normally distributed with mean 68.0 inches and standard deviation 3.0 inches. If one student is selected randomly, what is the probability that his/her height is
 - (a) greater than 72 inches.
 - (b) at least 65 inches.
 - (c) less than or equal to 64 inches.
 - (d) at most 80 inches.
 - (e) between 60 to 65 inches.
 - (f) above 65 inches and less than 71 inches.
 - (g) equal to 70 inches.
 - (h) How many students have heights greater than 74 inches but less than 80 inches.

Example (Solution)

- Let X be the height of students that follows normal distribution with mean $\mu = 68$ and standard deviation $\sigma = 3$

(a)

$$\begin{aligned}P(X > 72) &= P\left(\frac{X - \mu}{\sigma} > \frac{72 - 68}{3}\right) \\&= P(Z > 1.33) \\&= 1 - P(Z \leq 1.33) \\&= 1 - .9082 \\&= 0.0918\end{aligned}$$

Example (Solution)

(b)

$$\begin{aligned}P(X \geq 65) &= P\left(\frac{X - \mu}{\sigma} \geq \frac{65 - 68}{3}\right) = P(Z \geq -1.00) \\&= 1 - P(Z < -1.00) \\&= 1 - 0.1587 \\&= 0.8413\end{aligned}$$

Example (Solution)

(c)

$$\begin{aligned}P(X \leq 64) &= P\left(Z \leq \frac{64 - 68}{3}\right) \\&= P(Z \leq -1.33) \\&= 0.0918\end{aligned}$$

(d)

$$\begin{aligned}P(X \leq 80) &= P\left(Z \leq \frac{80 - 68}{3}\right) \\&= P(Z \leq 4.00) \\&= 1\end{aligned}$$

Example (solution)

(e)

$$\begin{aligned}P(60 < X < 65) &= P\left(\frac{60 - 68}{3} < \frac{X - \mu}{\sigma} < \frac{65 - 68}{3}\right) \\&= P(-2.67 < Z < -1.00) \\&= P(Z < -1.00) - P(Z < -2.67) \\&= 0.1587 - 0.0038 \\&= 0.1549\end{aligned}$$

Example (solution)

(f)

$$\begin{aligned}P(65 < X < 71) &= P\left(\frac{65 - 68}{3} < \frac{X - \mu}{\sigma} < \frac{71 - 68}{3}\right) \\&= P(-1.00 < Z < 1.00) \\&= P(Z < 1.00) - P(Z < -1.00) \\&= 0.8413 - 0.1587 \\&= 0.6826\end{aligned}$$

Example

(g) $P(X = 70) = 0$

(h)

$$\begin{aligned}P(74 < X < 80) &= P\left(\frac{74 - 68}{3} < \frac{X - \mu}{\sigma} < \frac{80 - 68}{3}\right) \\&= P(2.00 < Z < 4.00) \\&= P(Z < 4.00) - P(Z < 2.00) \\&= 1 - 0.9772 \\&= 0.0228\end{aligned}$$

Therefore, the expected number of students with heights greater than 74 inches but less than 80 inches is $300 \times 0.0228 = 6.84 \approx 7$

Exercise 1

- The lifetime of a single electric bulb have a mean of 720 hours and standard deviation 10 hours. We are assuming that the distribution of lifetime measured is normal. Compute the percentage of bulbs that have lifetime more than 740 hours.

Exercise 2

- Suppose that the height of UCLA female students has normal distribution with mean 62 inches and standard deviation 8 inches.
 - (a) Find the height below which is the shortest 30% of the female students.
 - (b) Find the height above which is the tallest 5% of the female students.

Thank You